

BikeReg

Predicting Urban Bicycle Flow in Mannheim



Ambition vs. Reality

- Mannheim is the birthplace of the bicycle¹ (Karl Drais, 1817)
- Mannheim wants to become a cycling city



- Current infrastructure does not reflect this ambition
- ADFC Fahrradklima-Test 2024: Mannheim achieved a school grade of **4.0**²
- **Major pain point:** Cyclists frequently stop at red lights due to car-centric signal timing (school grade **4.9**)



1: <https://www.mannheim.de/de/tourismus-entdecken/stadtgeschichte/stadtpunkte/buergertum-handel-industrie/karl-drais-und-seine-laufmaschine>

2: <https://mannheim.adfc.de/artikel/fahrradklima-test-2024-mannheim-stagniert-fahrradfreundlichkeit-ausbaufaehig>

Project Goal

- **Goal:** Develop a short-term forecasting model (1-hour horizon) to predict bicycle traffic volume
- **Application:** Enable dynamic traffic signal control (“Green Waves”) for cyclists
- **Inspiration:** Successful pilots in Amsterdam and Copenhagen where signals extend green phases for approaching platoons
- **Impact:** Reduce stops, increase safety and make cycling more attractive



Data Sources

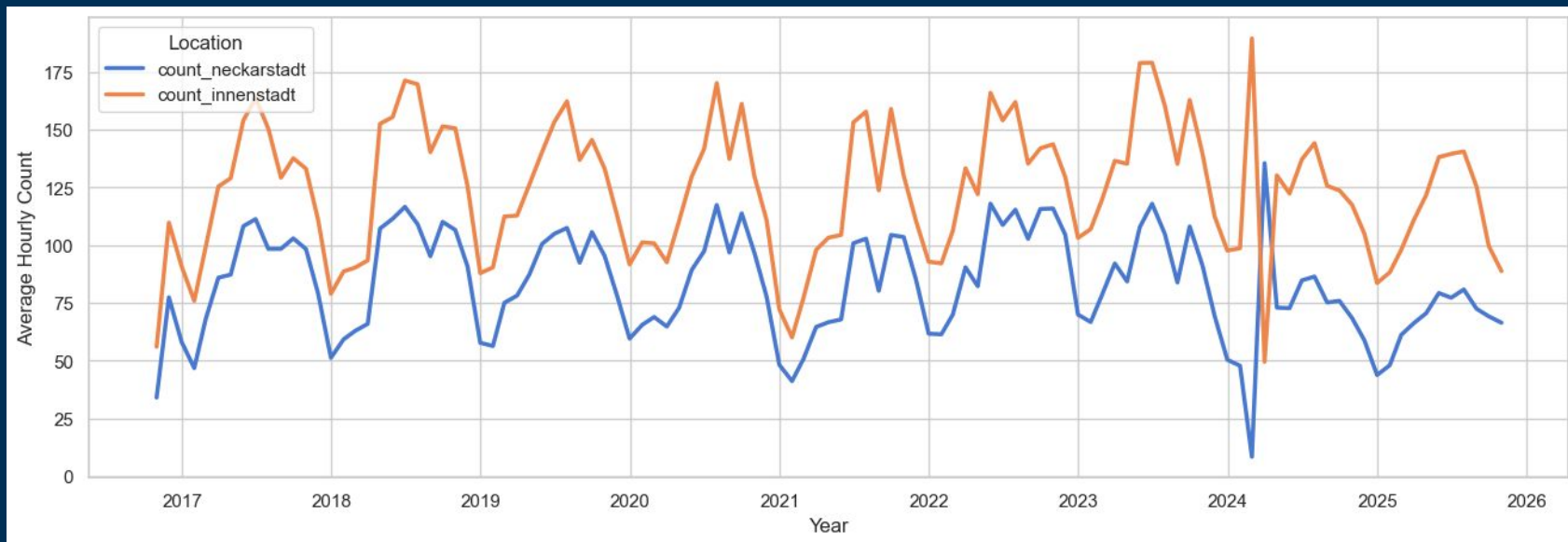


Data Sources

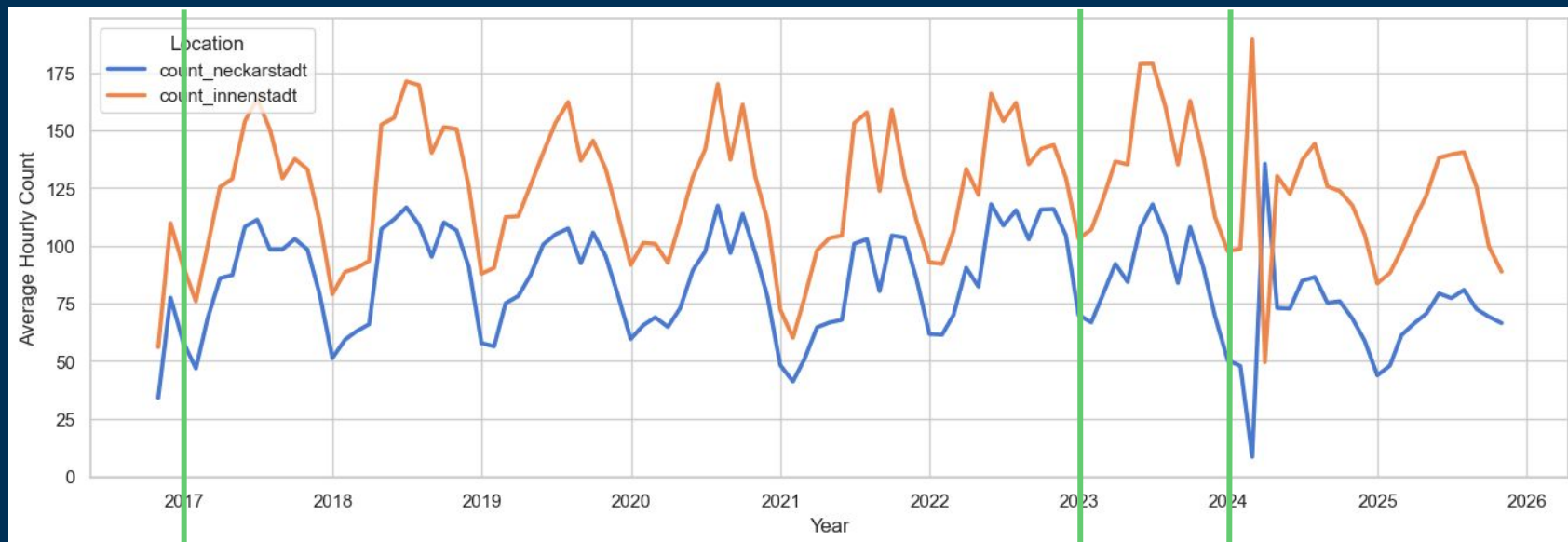
- **Primary source:** Kurpfalzbrücke Counters (Mannheim Open Data Portal)
 - 2 Counters, one for each side of the bridge
 - Hourly data of bikes crossing
 - Data from 2016 until now
- **External sources:**
 - **Weather:** Temperature, Precipitation (DWD API)
 - **Calendar:** Public & School Holidays (Open Holiday API), University Lecture times
 - **Events:** Public transit strikes (verdi/RNV)



Exploratory Data Analysis



Exploratory Data Analysis

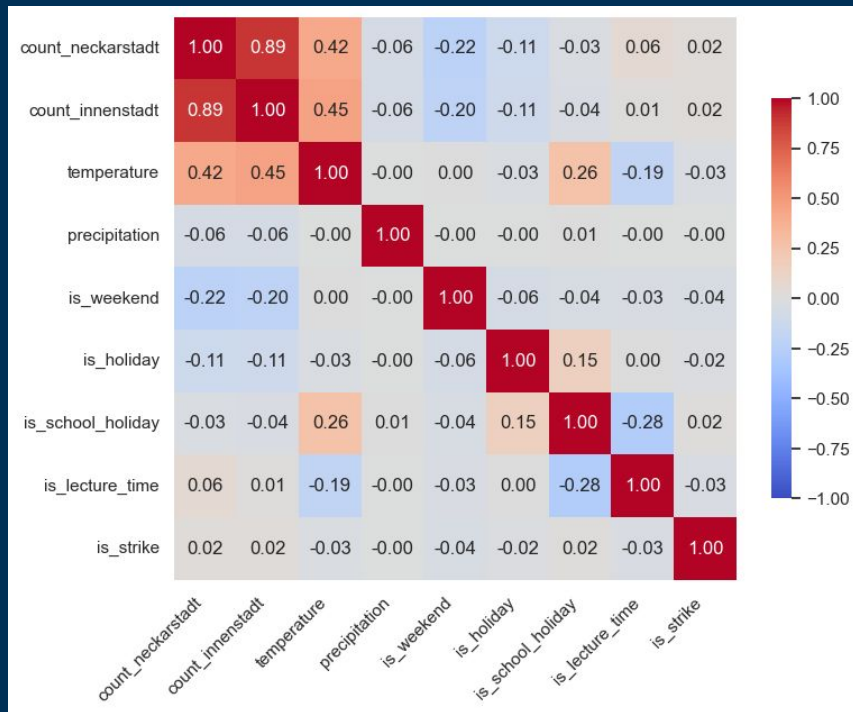


Train & Validation

Test



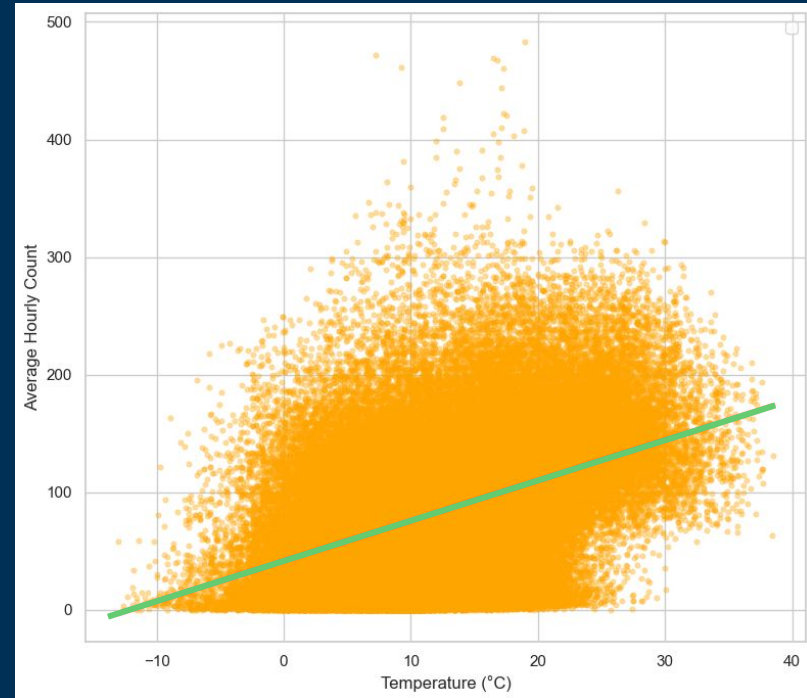
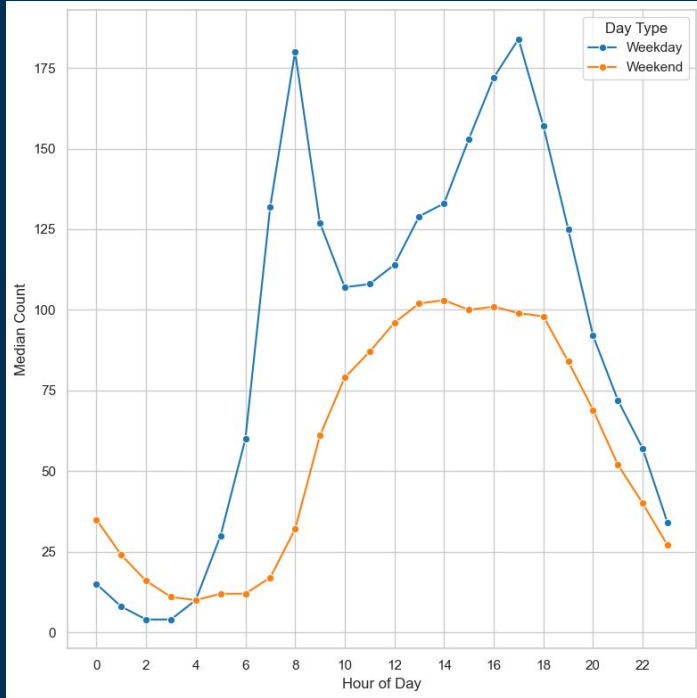
Exploratory Data Analysis



1. Both Counters highly correlate
2. **Weekend** is the most significant **negative** influence
3. **Temperature** is the most significant **positive** influence



Exploratory Data Analysis

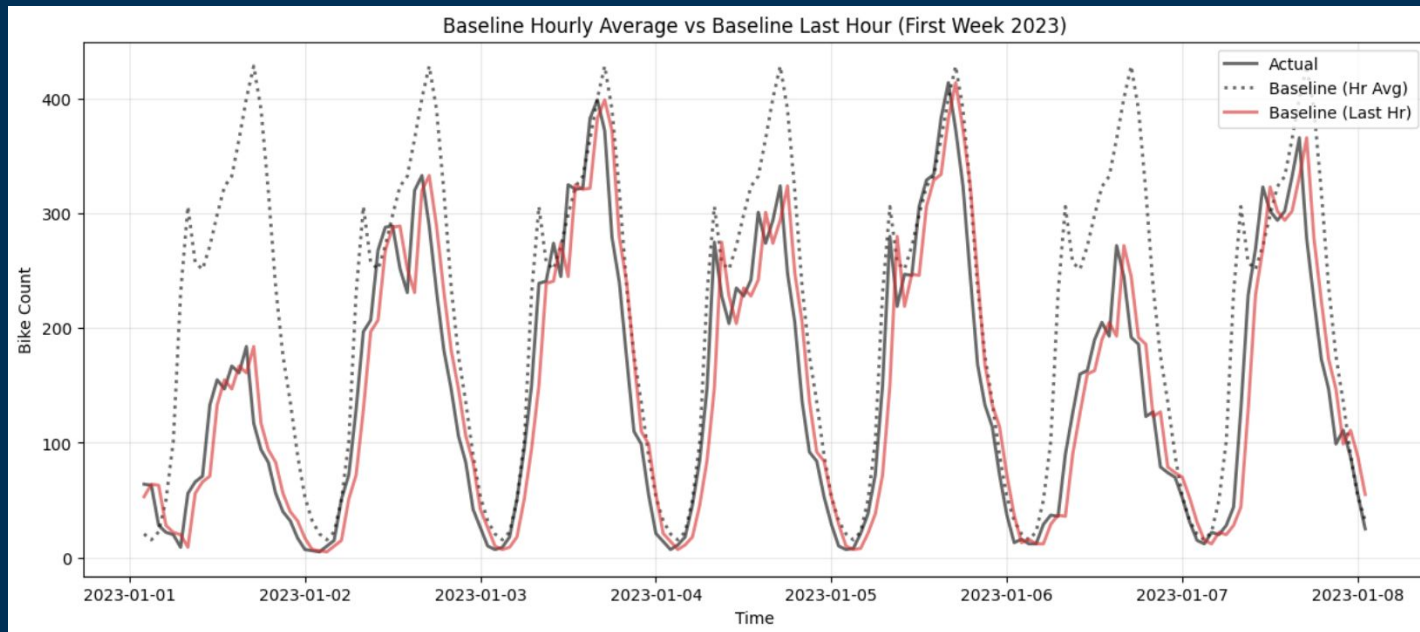


Preprocessing & Feature Engineering

- Merged both counters into one for total bridge throughput
- **Timeframe Restriction:** Limited to 2017-2023
- **Data Cleaning:**
 - Removed 59 rows where the target variables (counter values) were missing
 - Handled Daylight Saving Time (duplicate timestamps)
 - Imputed minor weather gaps (linear interpolation)
 - Removed major weather gaps with >5h missing data
- **Feature Engineering:**
 - Added binary flags for School/Public holidays, Strikes and Semester times
 - Binned hours into "Morning", "Midday", ... , "Late Night"
 - Derived meteorological seasons (Winter, Spring, Summer and Autumn)



Baselines & Lag



Hourly average:

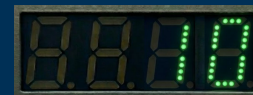
MAE: 70.38

R^2 : 0.67

Last Hour:

MAE: 52.29

R^2 : 0.83



Modeling Strategy

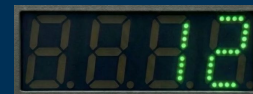
- **Validation:**
 - **Rolling-Origin Cross-Validation:** Time series aware -> K-Fold breaks time causality
- **Hyperparameter Optimization:**
 - **Optuna:** Bayesian Optimization to tune hyperparameters
- **Model Families Evaluated:**
 - **Regression** (Linear & Polynomial with Lasso & Ridge regularization)
 - **Autoregression** (Daily, Weekly and Bi-Weekly Sliding Window)
 - **Tree Ensembles** (Random Forest, XGBoost, LightGBM and CatBoost)



Evaluation & Error Analysis

Model	MAE (Count)	R^2 Score
<i>Baselines</i>		
Historical Hourly Avg	70.38	0.6756
Last Hour	52.29	0.8309
<i>Regression</i>		
Linear Ridge (No Lags)	77.49	0.6754
Linear Ridge ($\alpha = 9.33$)	39.45	0.8992
Linear Lasso ($\alpha = 0.48$)	39.37	0.8969
Polynomial Ridge ($d = 3, \alpha = 0.64$)	26.70	0.9533
Polynomial Lasso ($d = 3$)	26.80	0.9536
<i>Autoregressive</i>		
AR(24) - Daily	36.00	0.9155
AR(168) - Weekly	28.81	0.9462
AR(336) - Bi-Weekly	26.93	0.9514
<i>Ensemble Trees</i>		
Random Forest	24.45	0.9602
XGBoost	24.53	0.9600
LightGBM (Linear Tree)	24.40	0.9604
<u>CatBoost</u>	24.22	0.9610

- Linear Models performed the worst
- Autoregressive Models performed surprisingly well (with only time lag features)
- Winning Model: **CatBoost**
 - Error reduced by 53.7%
- Remaining Weaknesses:
 - Special Holidays (e.g. New Year's Eve)
 - Lag Shadowing



Conclusion & Future Work

Feasible:

- Forecast precision (MAE ~ 24) is sufficient to reliably predict traffic flow and support Green Waves

Future improvement:

- **Hybrid Control:** Use the model to set the Baseline Schedule, but use real-time sensors to override it during sudden spikes (solving Lag Shadowing)

Transferable:

- Model relies on temporal and weather data rather than specific infrastructure, allowing easy expansion to other bottlenecks like **Jungbuschbrücke**.





Thank you for your attention

Are there any questions?